



Full Length Article

Improved hydrodeoxygenation of bio-oil model compounds with polymethylhydrosiloxane by Brønsted acidic zeolites

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ABSTRACT

Bio-oil, one of significant renewable energy, was blocked from its direct application by thermodynamic instability due to the high oxygen content, so its deoxygenation needs urgent solution. Avoiding the disadvantages of the traditional catalytic method, such as harsh reaction conditions, potential explosion risk and environmental pollution, we utilize solid-acidic zeolites as low-load Pd-based catalyst supports instead of corrosive acid additives, with polymethylhydrosiloxane (PMHS) as the hydrogen-supplying agent, to achieve efficient hydrodeoxygenation of bio-oil models (carbonyl compounds) under mild conditions. The reaction conditions such as Si/Al ratio of HZSM-5 zeolite, temperature, solvent, and the type of Pd salts precursor are optimized. In particular, we have found that polar protic solvents improve catalytic efficiency by promoting proton transfer in the reaction. In an open-to-air, 97.9% ethylbenzene yield can be obtained for acetophenone conversion under mild conditions (0.5 wt% Pd/HZSM-5(18), 65 °C, 3 h, n-butanol as solvent), which is more efficient and environmental friendly than currently reports. Meanwhile, hydrogenation–dehydration mechanism was proposed, and the Brønsted acid in HZSM-5 promotes the dehydration of the alcohol (rate-limiting step) by efficiently accelerating the removal of hydroxyl groups and the proton transfer of the reaction. Furthermore, the catalytic scheme exhibits the excellent stability (reusable seven times) and versatility. The potential of a green catalytic technology using with PMHS opens attractive opportunities for bio-oil upgrading.

1. Introduction

Owing to the rapid economic development, fossil fuel depletion and global climate change were turning fierce, which prompts people to seek new renewable energy sources. Bio-oil, derived from fast biomass pyrolysis, is regarded as a promising second-generation renewable energy carrier [1,2]. Unfortunately, Bio-oil is a mixture with high oxygen content [3] and thermodynamic instability, including in ketones, aldehydes, alcohols, acids and esters et al., which hindered its direct application to existing devices [4,5]. Extensive researches have been carried out on the hydrodeoxygenation (HDO) for bio-oil upgrading [6,7].

Currently, several scholars have concentrated on bio-oil models to investigate the catalytic performance and mechanism. Among these processes, reductive deoxygenation of carbonyl compounds was severely useful for its widely employed transformation, and various potential methods have been developed [8]. Deoxygenation of ketone by the Wolff-kishner, Barton-McCombie [9] or Clemmensen reductions always accompanied by harsh reaction conditions and incomplete

transformation (e.g. ketone to alcohol), which was incompatible with the high selectivity of multifunctional compounds. Other alternative methods for reducing ketones to alkanes have been proposed, such as molecular hydrogen [10,11] or hydrides [12]. However, the main drawbacks associated with these reactions were both the high temperature or pressure reaction conditions and potential explosion risk.

It was well known that the reducing agent plays a critical role in the deoxygenation reaction. Compared to the hydrogen donors above (e.g. H₂), polymethylhydrosiloxane (PMHS) was an attractive option due to its inexpensive, non-toxic, stable to air and water, and more importantly, allows reduction reactions to occur under milder reaction conditions [13–16]. Unfortunately, few studies focus on heterogeneous catalytic systems for the deoxygenation of carbonyl compounds with PMHS [17–20]. Volkov et al. [17] used the heterogeneous Pd/C catalyst to deoxygenate aromatic ketones and aldehydes with PMHS as hydrogen donor under the mild condition. Whereas the limited surface area of supporter led to the Pd consumption increase and Economic degradation [18]. Catalytic efficiency of palladium can be considerably improved by

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